

4.5 Data Visualization & Types of Graphs

بار/کالم اقسام کا موازنہ، لائن وقت کا رجحان، پائی حصے، ہسٹوگرام مسلسل تقسیم (بار آپس میں لگے)، اسکیٹر دو متغیر...

CS XII

Chapter 4: Data and Analysis

Lecture 07 of 8

★ Golden

4.5 Data Visualization & Types of Graphs ★ Golden

LEARN IT

Pick the graph that matches the question. Then label axes and give a title — an unlabelled chart scores zero in an exam.

Chart	Shows	Watch out
Bar / column	Compare discrete categories (subjects, departments)	Column = vertical; bar often = horizontal
Line	Change over time (temperature, visitors)	Needs an ordered x-axis (months, years)
Pie	Parts of one whole as percentages	Bad with many or almost-equal slices
Histogram	Distribution of one numeric variable in bins	Bars touch (continuous range), unlike a bar chart
Scatter	Relationship of two numeric variables	Each dot is one pair (x, y)
Box plot	Min, Q1, median, Q3, max, and outliers	Best tool for spotting extreme values

4.5.1 Bar / column

```
import matplotlib.pyplot as plt
subjects = ["English", "Math", "Science", "History"]
marks = [64, 82, 71, 49]
plt.bar(subjects, marks, color="skyblue", edgecolor="black")
plt.title("Marks vs Subjects")
plt.xlabel("Subjects")
plt.ylabel("Marks out of 100")
```

4.5.2 Line

```
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
temps = [12, 15, 20, 25, 30, 35]
plt.plot(months, temps, marker="o")
plt.title("Average Monthly Temperature")
plt.grid(True)
```

4.5.3 Pie

```
subjects = ["Math", "Physics", "CS", "Biology"]
students = [25, 15, 30, 10] # total 80
plt.pie(students, labels=subjects, autopct="%1.1f%%", startangle=90)
```

Check: Math $25/80 = 31.25\% \rightarrow 31.2\%$. Physics 18.8% . CS 37.5% . Biology 12.5% .

4.5.4 Histogram

```
scores = [45, 50, 55, 60, 62, 65, 68, 70, 72, 75, 78, 80, 82, 85, 90, 95]
plt.hist(scores, bins=[40, 50, 60, 70, 80, 90, 100], edgecolor="black")
```

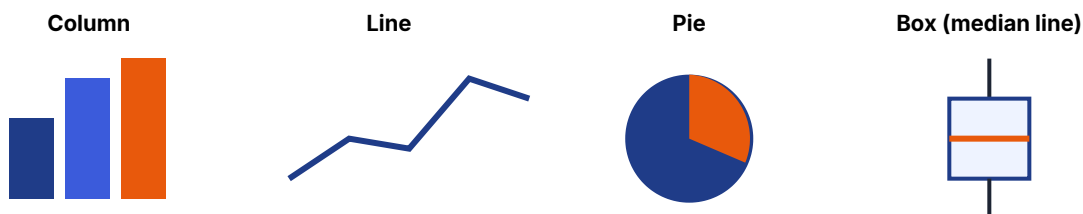
4.5.5 Scatter

```
plt.scatter([10, 20, 30, 40, 50, 60, 70, 80],
            [2, 4, 5, 4, 7, 8, 9, 12], color="green", alpha=0.7)
plt.title("Rainfall vs Crop Yield") # positive correlation
```

4.5.6 Box plot

```
plt.boxplot([[55, 60, 65, 70, 75, 80, 85],
            [40, 65, 70, 72, 75, 80, 95]],
            tick_labels=["Class A", "Class B"])
```

The box is the middle 50% (Q1 to Q3). The line inside is the median. Whiskers reach min and max (or a fence, with outliers drawn as dots). Class B is more spread out than Class A.



Column compares; line shows a trend; pie shows share; the box summarises spread around the median.

EXAM TIP

"Which graph and why?" is a 3-mark gift. Write the name, what it shows, and one reason the others are worse.

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CHECK YOURSELF

1. Why do histogram bars touch?
2. Rainfall vs yield — which chart?

Answers: 1. The x-axis is a continuous numeric range, not separate categories. 2. Scatter (two numeric variables, looking for correlation).

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Bar/column charts compare:

- a. Discrete categories
- b. Parts of one whole only
- c. A single time point of one var as a distribution
- d. A 5-number summary

MCQ 2. Line graphs are for:

- a. Trends over time
- b. Unrelated categories with gaps
- c. Shares of a budget
- d. Outliers in five numbers

MCQ 3. Pie charts need:

- a. Parts of one whole; few slices
- b. 60 slices of 1%
- c. Time on x
- d. Two numeric axes of correlation

MCQ 4. Histogram bars:

- a. Touch (continuous bins)
- b. Must have large gaps
- c. Are always time
- d. Replace the mean

MCQ 5. Scatter plots show:

- a. Two numeric variables (correlation)
- b. One category vs colour
- c. FIFO order
- d. OSI layers

MCQ 6. A box plot shows:

- a. Min, Q1, median, Q3, max (+ outliers)
- b. Only the mean
- c. Only mode
- d. Only n

Extra short questions

1. One-line job of each of the six charts.
2. Why histogram $\neq$ bar.
3. When pie fails.
4. Rainfall vs yield?

Extra long question

1. Six school questions $\rightarrow$ pick the chart and a matplotlib function (bar, plot, pie, hist, scatter, boxplot). Justify in one sentence each. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Bar categories; line time; pie share; hist distribution; scatter relationship; box spread/outliers.
2. Numeric bins, touching bars vs separate categories.
3. Too many / nearly equal slices.
4. Scatter; look for positive correlation.

Long

1. Enrolment by faculty: bar. Hits by month: plot. Club shares: pie. Heights: hist. Hours vs marks: scatter. Two classes' spread: boxplot. Unlabelled charts score 0.

Next lecture: 4.6 Descriptive Statistics